PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, GALLIUM ARSENIDE,
POWER RECTIFIER
TYPE 1N6751
JANTX, JANTXV and JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for Gallium arsenide rectifier diodes. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (similar to T0-257AA, isolated).

1.3 Maximum ratings.

Type	V_{RRM}	V_{RWM}	I_O 1/	I_{FSM} 2/	T_{STG}	T_{OP}	$R_{\theta JC}$	$R_{\theta JA}$
	V(pk)	V(PK)	A	A(pk)	°C	°C	°C/W	°C/W
1N6751	180	180	9	55	-55 TO +200	-55 TO +200	5.4	65

1/ Derate at .162 a/°C above Tc = 95°C.
2/ Surge applied at rated load conditions, half wave, 1ϕ 60 Hz.
3/ Total package current is limited to 20 amps.

1.4 Primary electrical characteristics. Primary electrical characteristics at 25°C unless otherwise specified.

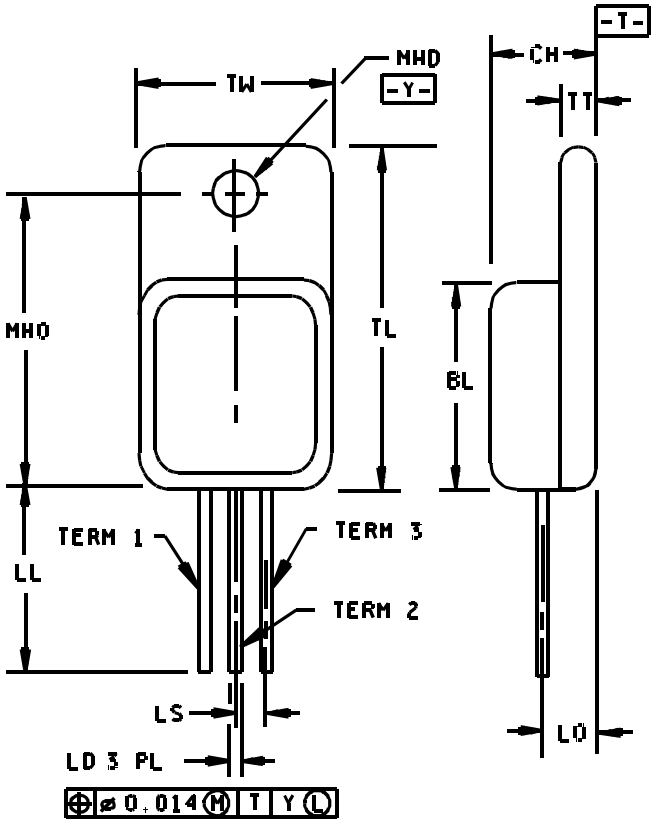
Type	V_{FM1} I_{FM} = I_{O} A	V_{FM2} I_F = 5 A	I_{R1} V_R = 90V	I_{R2} V_R = V_{RWM}	I_{R3} V_R = 90V T_A = 125 °C	I_{R4} V_R = V_{RWM} T_A = 125 °C	I_{RM} V_R=150V I_F = 5 A	t_{rr} V_R=5V	C_{T1} di/dt = 200 A/µs
	V dc	V dc	µA dc	µA dc	µA dc	µA dc	A	ns	pF
1N6751	1.55	1.20	1.0	25	120	685	1.6	13.0	300

* Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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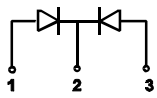
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Dimensions				
Symbol	Millimeters		Inches	
	Min	Max	Min	Max
BL	10.4	10.9	0.410	0.430
CH	4.82	5.10	0.190	0.200
LD	0.63	0.90	0.025	0.035
LO	3.05 typ		0.120 typ	
LS	2.50 bsc		0.100 bsc	
MHD	3.55	3.80	0.140	0.150
MHO	13.4	13.6	0.527	0.537
TL	16.4	16.9	0.645	0.665
TT	0.90	1.15	0.035	0.045
TW	10.4	10.7	0.410	0.420



Configuration

Terminal	Description
1	Anode
2	Cathode
3	Anode
4	No Connection



Schematic diagram

NOTES:
1. Dimensions are Inches in millimeters equivalents are given for general information only See paragraph 3.3.
2. All terminals are isolated from case.
3. Terminal 4 is the case

FIGURE 1. Physical dimensions (T0-257AA).

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figure 1 (T0-254AA) herein. Methods used for electrical isolation of the terminal feedthroughs shall employ materials that contain a minimum of 90% AL_2O_3(ceramic). Examples of such construction techniques are metallized ceramic eyelets or ceramic walled packages. The preferred measurements used herein are the metric units. However, this diode was designed using inch-pound units of measurement. In case of conflicts between the metric and inch-pound units, the inch-pound units shall be ruled.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in paragraphs 4.4.2 and 4.4.3.

3.7 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

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4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified in table III herein.

4.3 Screening (JANS, JANTX, AND JANTXV levels only). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3C 1/	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
9	I_{R1}, V_{F1}, and $V_{(BR)}$	Not applicable
11	I_{R1}, V_{F1}, $V_{(BR)}$; $\Delta I_{R1} \leq$ 100 percent of initial value or ±15 nA, whichever is greater. $\Delta V_{F1} \leq$ ±0.02 V.	I_{R1} and V_{F1}
12	See 4.3.1	See 4.3.1
13 2/	Subgroup 2 and 3 of table I herein; $\Delta I_R \leq$ 100 percent of initial value or 15 nA, whichever is greater. $\Delta V_{BR} \leq$ ±15 percent of initial reading.	Subgroup 2 of table I herein; $\Delta I_{R1} \leq$ 100 percent of initial value or 15 nA dc, whichever is greater. $\Delta V_{F1} \leq$ 0.02 V dc.

1/ Thermal impedance shall be performed any time after screen 3.
2/ $Z_{\Theta JX}$ is not required in screen 13 if already performed.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

MIL-STD-750 method 1038, test condition A. T_C = +125°C; V_R = 80% or rated V_{RRM} dc.

4.3.2 Thermal impedance $Z_{\Theta JX}$ measurements for screening. The $Z_{\Theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit (not to exceed the group A, subgroup 2 limit) for $Z_{\Theta JX}$ in screening (table IV of MIL-PRF-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for engineering evaluation and disposition.

4.3.3 Thermal impedance ($Z_{\Theta JX}$ measurements) for initial qualification or requalification. The $Z_{\Theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101 (read and record data $Z_{\Theta JX}$) derived conditions limits and thermal response curve shall be supplied to the qualifying activity on the qualification lot prior to qualification approval.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein. The following test conditions shall be used for $Z_{\Theta JX}$, group A inspection: $Z_{\Theta JX} \leq$ 3.0 ° C/W

I_H = 15 - 30 A

t_H = 50 ms

I_M = 10 mA

t_{MD} = 100 ms minimum

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JANTX and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
3	4066	T_A = +25°C ± 5°C. See Table 1 subgroup 6 herein.

4.4.2.1 Group B inspection, table VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
3	4066	T_A = +25°C ±5°C See Table 1 subgroup 6 herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

Subgroup	Method	Condition
2	2036	Test condition A, 5 pounds, t = 15 s ± 3 s.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps and footnotes of table II herein.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	2/ Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3101	See 4.3.2	$Z_{\Theta JX}$		3.0	°C/W
Forward voltage	4011	Pulse test see note 2/				
1N6751		I_{FM} = 9A	V_{FM1}		1.55	V dc
		I_{FM} = 5A	V_{FM2}		1.20	V dc
Reverse current	4016	DC method;				
1N6751		V_R = 90V dc	I_{R1}		1.0	µA dc
		V_R = 180 V dc	I_{R2}		25	µA dc
Subgroup 3						
High temperature operation		T_A = +125°C				
Reverse current	4016	DC method;				
1N6751		V_R = 90 V dc	I_{R3}		120	µA dc
		V_R = 180 V dc	I_{R4}		685	µA dc
Low-temperature operation		T_A = -55°C				
Forward voltage	4011	Pulse test. See note 2/ I_{FM3} = A dc	V_{F4}			
1N6751					1.55	V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	2/ Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Capacitance	4001	V_R = 5 V dc; V_{sig} = 50 mV(p-p) f = 1 MHz	C_{T1}			
1N6751					150	pF
Reverse recovery time	4031	Condition A V_R = 150 V di/dt = 200 A/µs	t_{rr}			
1N6751					13.0	ns
Subgroup 5						
Not applicable						
Subgroup 6						
Surge current	4066	I_{FSM} = 60 A(pk) V_{RM} = 180 V(pk) I_O = 9 A(pk) 10 surges of 8.3 mS each at 1 minnute intervals				
Electrical measurements		See table II herein.				
Subgroup 7						
Not applicable						

1/ For sampling plans not specified, see MIL-PRF-19500.

2/ Pulse test: Pulse width = 300µS, duty cycle ≤ 2%

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TABLE II. Groups A, B, C and E electrical measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Forward voltage	4011	t_p = 5 ms I_F = 10 mA dc pulsed	ΔV_{F1} 4/		±25 mV dc maximum change.	
2	Reverse current	4016	DC method V_R = 20 V dc	ΔI_{R1} 4/		100 percent or 20 nA dc which- ever is greater.	

1/ The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows: Subgroup 3, 4, and 5, see table II herein, steps 1 and 2.

2/ The electrical measurements for table VIb (JANTX and JANTXV) of MIL-PRF-19500 are as follows: Subgroups 2, 3, and 6, see table II herein, steps 1 and 2.

3/ The electrical measurements for table VII of MIL-PRF-19500 are as follows: Subgroup 2, 3, and 6, see table II herein, steps 1 and 2 for JANS, JANTX, and JANTXV.

4/ Devices which exceed the group A limits for this test shall not be accepted.

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TABLE III. Group E inspection (all quality levels) for qualification only. 1/

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			22 devices c = 0
Thermal shock (temperature cycling)	1051	500 cycles, -55, +200°C	
Hermetic seal	1071		
Fine Leak		Test condition H 5 x 10^{-7} atm cc/s	
Gross Leak		Test Condition C or K	
Electrical measurement		See table II herein	
Subgroup 2			22 devices c = 0
Steady-state reverse bias	1038	test condition A; 1,000 hours, See 4.3, screen 10	
Electrical measurement		See table II herein	
Subgroup 3			3 devices c = 0
Destructive Physical Analysis			
Subgroup 4			22 devices c = 0
Thermal resistance,	3101	$Z_{\Theta JX} \leq 4.0$°C/W I_H = 15 - 30 A t_H = 50 ms I_M = 10 mA t_{MD} = 500 ms minimum	
Subgroup 5			c = 0
Not applicable			
Subgroup 6			
Not applicable			

1/ For initial design and process change verification only (one time testing).

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish or formation may be specified (see 3.3.1).

c. Type designation and product assurance level.

6.3 Cross reference list. Parts from this specification may be used to replace the following commercial Part of Identifying Numbers (PIN):

Preferred types	Commercial types
JANTX1N6751	1N6751

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - ES
Air Force - 17
NASA - NA

Preparing activity:
DLA - ES

(Project 5961-1855)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/636	**2. DOCUMENT DATE (YYMMDD)** 96/10/21

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, DIODE, GALLIUM ARSENIDE, POWER RECTIFIER TYPE 1N6751 JANTX, JANTXV and JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 614-692-0510 (2) AUTOVON 850-0516
c. ADDRESS (Include Zip Code) Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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